10 **(a)**	energy required to separate the nucleons (in a nucleus)	M1	
	to infinity	A1	[2]
	(*allow reverse statement*)		
(b) **(i)**	$\Delta m = (2 \times 1.00867) + 1.00728 - 3.01551$	C1	
	$= 9.11 \times 10^{-3}$ u	C1	
	binding energy $= 9.11 \times 10^{-3} \times 930$		
	$= 8.47$ MeV	A1	[3]
	(*allow 930 to 934 MeV so answer could be in range 8.47 to 8.51 MeV)*		
	(*allow 2 s.f.*)		
(ii)	$\Delta m = 211.70394 - 209.93722$		
	$= 1.76672$ u	C1	
	binding energy per nucleon $= (1.76672 \times 930)/210$	C1	
	$= 7.82$ MeV	A1	[3]
	(*allow 930 to 934 MeV so answer could be in range 7.82 to 7.86 MeV*)		
	(*allow 2 s.f.*)		
(c)	total binding energy of barium and krypton	M1	
	is greater than binding energy of uranium	A1	[2]

Section B